

MATH 231 – Linear Algebra I

The Vector Calculus Unit, Part 1 ■ Derivatives, and the Gradient in the Plane

Kiely Hall 326 ■ Instructor: Kevin Bedoya

About this document. This opens a unit on calculus in several variables. It begins with a review of one-variable differentiation — material from MATH 141 — not as a courtesy, but because the central definition of §6 turns out to be the one-variable definition with a single substitution made inside it. Having that definition fresh, and in the same notation, is what makes the new one read as a small change rather than a new idea.

Everything after §4 concerns functions of *two* variables, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. That is not the general case, and it is not meant to be. Two variables is where every idea in the subject can be drawn on a page, and the plane is the space the vector unit has been building all semester. Part 2 of this unit lifts the whole of §5–§9 to $\mathbb{R}^n$, where the pictures stop and the formulas do not change.

A common confusion (Section numbers restart). This unit numbers its sections from 1 again. They do *not* continue from the vector unit (§1–§24), the computational unit, or the matrix unit, all of which also start at 1. A bare “§5” below means §5 *of this unit*; whenever we point at another we will name it. Part 2 of this unit continues the numbering from §10, so a reference to §12 there is not a reference to the vector unit’s §12.

Assumed. From MATH 141: limits, and differentiation of one-variable functions. From the vector unit — all of it from lectures already given — $\mathbb{R}^2$ and its vectors (§1, §5); the Euclidean norm (§2); function notation $f : A \rightarrow B$ (§6); the dot product and the identity $\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$ (§11.3); the orthogonality test (§11.7); and normalisation into a unit vector (§12.1). Nothing from vectors Part 4 or Part 5 is used, and neither is the matrix unit.

What you should be able to do afterwards. State the definition of the derivative and use the standard rules; state the algebraic properties of e^x and $\ln x$ and explain why the two lists mirror each other; derive $\frac{d}{dx} \ln x$ from the definition and $\frac{d}{dx} e^x$ two ways; read the signature $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and give examples; define the two partial derivatives as limits, compute them, and say what they are on the surface $z = f(x, y)$; say what kind of object a partial derivative is; assemble the partials into the gradient; define the directional derivative and compute it as $\langle \nabla f, \mathbf{u} \rangle$; show that the rate of change is greatest along the gradient and explain why that makes ∇f the direction of steepest ascent; say what $-\nabla f$ is and how gradient descent uses it; and write the linear approximation of a function of one variable and of two.

1. The derivative, in one variable

1.1. The definition

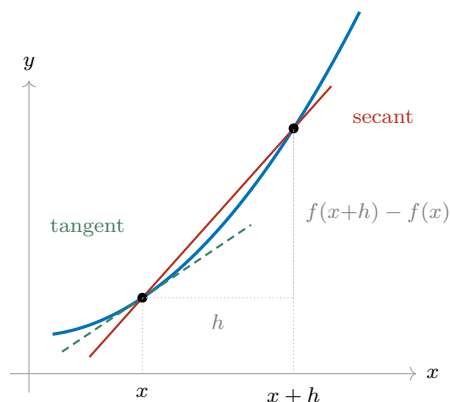
Definition 1.1 (Derivative). Let $f : \mathbb{R} \rightarrow \mathbb{R}$. The **derivative** of f at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad (1)$$

provided the limit exists. When it does, f is called **differentiable** at x .

The quotient inside the limit is the slope of the line through the two points $(x, f(x))$ and $(x+h, f(x+h))$ — a *secant* line. As h shrinks, the second point slides toward the first and the secant pivots.

The derivative is the slope it settles on: the slope of the tangent line at x , and the instantaneous rate at which f changes there.



1.2. Notation

The same object is written several ways, and all of them appear in practice:

$$f'(x), \quad \frac{df}{dx}, \quad \frac{dy}{dx}, \quad \frac{d}{dx}f(x).$$

The last form is useful because $\frac{d}{dx}$ reads as an *instruction* — “differentiate what follows with respect to x ” — which is how it is used in §4.2.

2. The rules

Computing every derivative from (1) would be unbearable, so the limit is evaluated once for each basic function and once for each way of combining functions. What follows is the resulting catalogue. Throughout, c is a constant and f, g are differentiable.

2.1. Structural rules

These say how differentiation interacts with the ways functions are built out of other functions.

Rule	Statement	Notes
Constant	$\frac{d}{dx} c = 0$	a flat graph has slope 0
Constant multiple	$\frac{d}{dx} [c f(x)] = c f'(x)$	scalars pass through
Sum / difference	$\frac{d}{dx} [f(x) \pm g(x)] = f'(x) \pm g'(x)$	term by term
Power	$\frac{d}{dx} x^n = n x^{n-1}$	any real n
Product	$\frac{d}{dx} [f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$	<i>not</i> $f'g'$
Quotient	$\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$	order of the numerator matters
Chain	$\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$	outer at inner, times inner

The first three together say that differentiation is *linear*: it distributes across sums and lets constants through. That word is worth registering — it is the same property, in the same sense, that the vector unit attached to other operations, and it will return when the gradient appears.

2.2. Trigonometric functions

Function	Derivative	Function	Derivative
$\sin x$	$\cos x$	$\csc x$	$-\csc x \cot x$
$\cos x$	$-\sin x$	$\sec x$	$\sec x \tan x$
$\tan x$	$\sec^2 x$	$\cot x$	$-\csc^2 x$

A pattern worth using as a memory aid: every co-function — $\cos$, $\csc$, $\cot$ — picks up a minus sign. The exponential and the logarithm are missing from these tables on purpose. Their derivatives are not quoted here; they are derived in §4.

3. The exponential and the logarithm

Before differentiating these two, their algebra is worth setting out — partly because §4.1 uses it, and partly because the two lists are the same list read in a mirror.

3.1. The exponential function

The function $\exp(x) = e^x$ has domain $\mathbb{R}$ and range $(0, \infty)$: it is *never* zero or negative, no matter what is fed to it. It is strictly increasing.

	Property	In words
E1	$e^0 = 1$	the exponential of nothing is one
E2	$e^{a+b} = e^a e^b$	sums become products
E3	$e^{a-b} = \frac{e^a}{e^b}$	differences become quotients
E4	$(e^a)^b = e^{ab}$	powers multiply the exponents
E5	$e^{-a} = \frac{1}{e^a}$	negation inverts

E2 is the essential one; E3 and E5 are consequences of it.

3.2. The natural logarithm

The function $\ln x$ has domain $(0, \infty)$ and range $\mathbb{R}$ — exactly the exponential's domain and range, exchanged. That is the first sign that the two are inverses.

	Property	In words
L1	$\ln 1 = 0$	mirrors E1
L2	$\ln(ab) = \ln a + \ln b$	products become sums
L3	$\ln \frac{a}{b} = \ln a - \ln b$	quotients become differences
L4	$\ln(a^r) = r \ln a$	exponents come down in front
L5	$\ln e = 1$	the base is its own unit

3.3. They are inverses

The two functions undo each other:

$$\ln(e^x) = x \quad \text{for all } x \in \mathbb{R}, \quad e^{\ln x} = x \quad \text{for all } x > 0. \quad (2)$$

This is what explains the parallel between the two tables. Read E1 through E4 backwards and they become L1 through L4: E2 says e turns addition into multiplication, so its inverse must turn multiplication back into addition, which is L2. The logarithm laws are not a second set of facts to memorise — they are the exponential laws seen from the other side.

A common confusion ($\ln$ of a sum). L2 converts the log of a *product* into a sum. There is no corresponding rule for the log of a sum: $\ln(a + b)$ does not simplify, and in particular it is not $\ln a + \ln b$. Check it once with $a = b = 1$: $\ln 2 \approx 0.693$, while $\ln 1 + \ln 1 = 0$.

4. Deriving the two derivatives

These two derivatives were deliberately left out of the tables in §2. Here they are earned instead. The order matters: the logarithm is done first, from the definition alone, and the exponential then follows from it.

4.1. The natural logarithm, from the definition

Let $f(x) = \ln x$ with $x > 0$. Apply (1) directly:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x}{h}.$$

The numerator is a difference of logarithms, so L3 folds it into one:

$$= \lim_{h \rightarrow 0} \frac{1}{h} \ln\left(\frac{x+h}{x}\right) = \lim_{h \rightarrow 0} \frac{1}{h} \ln\left(1 + \frac{h}{x}\right).$$

Now use L4 in reverse — a coefficient in front of a logarithm can be pushed up into the exponent — moving the $1/h$ inside:

$$= \lim_{h \rightarrow 0} \ln\left[\left(1 + \frac{h}{x}\right)^{1/h}\right] = \ln\left[\lim_{h \rightarrow 0} \left(1 + \frac{h}{x}\right)^{1/h}\right].$$

Moving the limit inside $\ln$ is legitimate because $\ln$ is continuous, and a continuous function may be evaluated at the limit of its input.

The remaining limit is a standard one. Substituting $u = h/x$, so that $u \rightarrow 0$ as $h \rightarrow 0$ and $1/h = 1/(ux)$,

$$\lim_{h \rightarrow 0} \left(1 + \frac{h}{x}\right)^{1/h} = \lim_{u \rightarrow 0} \left[(1 + u)^{1/u}\right]^{1/x} = e^{1/x},$$

using $\lim_{u \rightarrow 0} (1 + u)^{1/u} = e$, which is one of the standard definitions of e . Therefore

$$f'(x) = \ln(e^{1/x}) = \frac{1}{x},$$

the last step by (2). ■

4.2. The exponential, by implicit differentiation

With $\frac{d}{dx} \ln x = 1/x$ in hand, the exponential costs three lines. Let

$$y = e^x.$$

Take the logarithm of both sides, which by (2) gives

$$\ln y = x.$$

Differentiate both sides with respect to x . The right side gives 1. The left side is a function of y , which is itself a function of x , so the chain rule applies:

$$\frac{d}{dx} [\ln y] = \frac{d}{dx} [x] \quad \implies \quad \frac{1}{y} \frac{dy}{dx} = 1.$$

Solving for the derivative and substituting back,

$$\frac{dy}{dx} = y = e^x. \quad \blacksquare$$

This technique — differentiating an equation in which y is not isolated, then solving for dy/dx — is **implicit differentiation**. It works here because writing $\ln y = x$ converts a hard derivative into an easy one at the cost of an extra chain-rule factor.

4.3. The exponential, from the definition

The same result comes out of (1). Let $f(x) = e^x$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h}.$$

By E2, $e^{x+h} = e^x e^h$, so e^x is a common factor of the numerator:

$$= \lim_{h \rightarrow 0} \frac{e^x e^h - e^x}{h} = \lim_{h \rightarrow 0} \frac{e^x (e^h - 1)}{h} = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h}.$$

The factor e^x does not involve h , so it passes outside the limit. What remains is the standard identity

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1, \tag{3}$$

which gives $f'(x) = e^x$. ■

Remark. It is worth being clear about what this second derivation does and does not establish. Setting $x = 0$ in (1) for $f(x) = e^x$ turns it into exactly (3) — so the identity *is* the statement that $f'(0) = 1$. The argument therefore does not conjure the derivative from nothing; it shows that the derivative everywhere follows from its value at the single point $x = 0$. That is a real statement, and a strong one: the exponential is pinned down completely by its slope at the origin. The derivation in §4.2 is the one that stands independently, resting on §4.1.

5. Functions of two variables

5.1. The signature

Everything so far has concerned $f : \mathbb{R} \rightarrow \mathbb{R}$ — one number in, one number out. Almost nothing worth computing has that shape.

From the vector unit. The notation $f : A \rightarrow B$ was fixed in §6 of the vector unit: the set before the arrow is what the function consumes, the set after it is what the function produces. That notation now does real work.

Definition 5.1 (Scalar function of two variables). A **scalar function of two variables** is a function

$$f : \mathbb{R}^2 \longrightarrow \mathbb{R},$$

taking a vector $\mathbf{x} = [x, y] \in \mathbb{R}^2$ and returning a single real number $f(\mathbf{x})$, also written $f(x, y)$.

The input is a *vector*; the output is a *scalar*. That asymmetry is the whole subject of the rest of this document. Such functions are also called **scalar fields**: they assign one number to every point of the plane.

Remark (Two notations, one function). $f(\mathbf{x})$ and $f(x, y)$ name the same thing. The first emphasises that the input is a single object — a point of $\mathbb{R}^2$ — and is the form to keep in mind when the vector operations of the vector unit are applied to it. The second is what you write when you actually differentiate. Both appear below, and switching between them is not a change of subject.

5.2. Examples

Example 5.1. Each of the following is a function $\mathbb{R}^2 \rightarrow \mathbb{R}$.

- (a) $f(x, y) = x^2 + y^2$. It returns the squared distance from the origin. $f(3, 4) = 25$.
- (b) $f(x, y) = xy$. The area of a rectangle with the given side lengths. $f(3, 4) = 12$.
- (c) $f(\mathbf{x}) = \|\mathbf{x}\| = \sqrt{x^2 + y^2}$. The *Euclidean norm* is a function $\mathbb{R}^2 \rightarrow \mathbb{R}$ — it eats a vector and returns its length, a single number. You have been using an example of this kind of function since the vector unit's §2.
- (d) $f(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle = a_1x + a_2y$ for a fixed vector $\mathbf{a} = [a_1, a_2]$. The dot product against a fixed vector is likewise $\mathbb{R}^2 \rightarrow \mathbb{R}$ (vector unit §11.3).
- (e) $f(x, y) = e^{-(x^2+y^2)}$, a bump peaking at the origin and decaying in every direction.
- (f) The temperature at each point of a flat plate, or the elevation at each point on a map: both assign one number to each point of a two-dimensional region.

Remark. Examples (c) and (d) are worth a second look. Two of the most-used objects from the vector unit — the norm and the dot product — are exactly the kind of function this section is about. Nothing new was introduced; they were simply never described this way before.

Remark (Picturing one). A function $\mathbb{R}^2 \rightarrow \mathbb{R}$ has a graph in three dimensions: over each point (x, y) of the plane, plot the height $z = f(x, y)$. The result is a *surface*. Example (a) is a bowl, example (e) is a hill, and example (f) is the literal landscape the word “elevation” suggests. This picture is worth holding on to: almost every geometric statement below is a statement about walking around on that surface.

5.3. What “the derivative” should mean here

Now try to differentiate one of these, and the definition (1) immediately fails to typecheck. It reads

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h},$$

and if the input is a vector then “ $x + h$ ” is a vector plus a number, which is undefined. Even if it were repaired, a deeper problem remains: for a one-variable function there are only two ways to leave the point x — left or right — and the two are related by a sign. In the plane there are infinitely many directions to leave $\mathbf{x}$ in, and f may change at a different rate along each one.

Walking east from a point on a hillside might take you uphill; walking north from the same point might take you down. “The rate of change of the elevation” is not yet a well-posed question. It becomes one only after a direction is named.

The next section names the two simplest ones. §8 names all the rest.

6. Partial derivatives

6.1. The definition

The simplest directions to move in are the two coordinate directions: change x and leave y alone, or change y and leave x alone. Each choice turns the two-variable function into a one-variable one, and (1) applies to that.

Definition 6.1 (Partial derivatives). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. The **partial derivative of f with respect to x** at the point (x, y) is

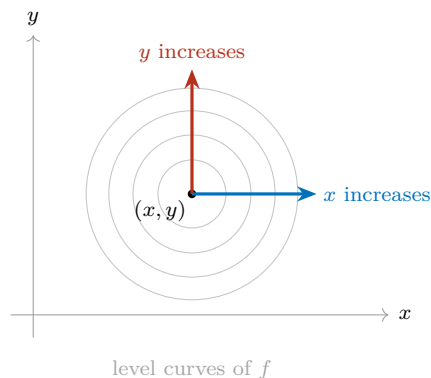
$$\frac{\partial f}{\partial x}(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}, \quad (4)$$

and the **partial derivative of f with respect to y** is

$$\frac{\partial f}{\partial y}(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}, \quad (5)$$

each provided the limit exists.

Compare (4) with (1). They are the same limit. In the first, the step h is added to x while y never moves; in the second, h is added to y while x never moves. The curly ∂ replaces d to record that there were other directions available and this one was chosen.



Each arrow is one partial derivative: step off the point along one coordinate axis, and measure the rate at which f responds.

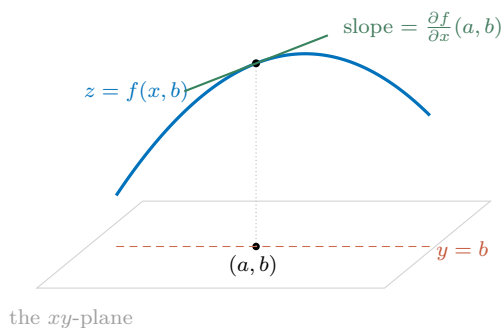
6.2. What they look like on the surface

The picture above is drawn in the plane, where the function lives as a set of level curves. The more telling picture is drawn on the graph.

Fix the point (a, b) and hold y at the value b . The set of points with $y = b$ is a line in the plane; sitting above it on the surface $z = f(x, y)$ is a single curve — the **slice** of the surface cut by the vertical plane $y = b$. Along that curve only x varies, so it is the graph of the one-variable function

$$g(x) = f(x, b).$$

And $\partial f / \partial x(a, b)$ is exactly $g'(a)$: the ordinary slope of the tangent line to that slice, at that point.



Holding x at a instead cuts the surface with the plane $x = a$, producing a different curve, and $\partial f / \partial y(a, b)$ is the slope of *its* tangent at the same point. Two planes through one point of the surface, two curves, two slopes.

Remark. This is why a single number cannot be “the slope” of a surface. The surface has a different slope in every direction you slice it, and the two partial derivatives report only the two slices lined up with the axes. §8 asks what happens along all the other slices.

6.3. Computing them

In practice the limit is never evaluated. Since the other variable is held fixed, it behaves as a constant, and the expression becomes a function of the one live variable — so the rules of §2 apply unchanged.

To compute $\partial f/\partial x$: treat y as a constant, and differentiate as in one-variable calculus.
 To compute $\partial f/\partial y$: treat x as a constant and do the same.

Example 6.1. Let $f(x, y) = x^2y + 3y^2$.

For $\partial f/\partial x$, treat y as a constant. Then x^2y is a constant times x^2 , and $3y^2$ is a constant, whose derivative is zero:

$$\frac{\partial f}{\partial x} = 2xy.$$

For $\partial f/\partial y$, treat x as a constant. Now x^2y is a constant times y , and $3y^2$ is live:

$$\frac{\partial f}{\partial y} = x^2 + 6y.$$

Example 6.2. Let $f(x, y) = e^{-(x^2+y^2)}$, example (e) of §5.2. By the chain rule, holding y fixed,

$$\frac{\partial f}{\partial x} = e^{-(x^2+y^2)} \cdot (-2x) = -2x e^{-(x^2+y^2)}, \quad \frac{\partial f}{\partial y} = -2y e^{-(x^2+y^2)}.$$

The two are the same computation with the roles of x and y exchanged, which they must be: the formula for f is unchanged by that exchange.

Example 6.3. Let $f(x, y) = x^3 \ln y + \frac{y}{x}$, for $x \neq 0$ and $y > 0$. Holding y fixed, $\ln y$ is a constant multiplier and $y/x = y \cdot x^{-1}$:

$$\frac{\partial f}{\partial x} = 3x^2 \ln y - \frac{y}{x^2}.$$

Holding x fixed, x^3 is a constant multiplier, $\frac{d}{dy} \ln y = 1/y$ by §4.1, and y/x is a constant times y :

$$\frac{\partial f}{\partial y} = \frac{x^3}{y} + \frac{1}{x}.$$

Example 6.4. Let $f(x, y) = \sin(xy)$. Holding y fixed, the inside function is xy , whose derivative with respect to x is y ; the chain rule gives

$$\frac{\partial f}{\partial x} = y \cos(xy), \quad \text{and symmetrically} \quad \frac{\partial f}{\partial y} = x \cos(xy).$$

Note that the held variable does not disappear — it survives as a factor, because it was a coefficient of the live variable inside the sine.

Example 6.5. Let $f(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle = a_1x + a_2y$, example (d) of §5.2, with $\mathbf{a} = [a_1, a_2]$ fixed. Only the first term involves x , and in it x appears to the first power with coefficient a_1 ; the second term is a constant. So

$$\frac{\partial f}{\partial x} = a_1, \quad \frac{\partial f}{\partial y} = a_2.$$

The partial derivatives of a dot product against $\mathbf{a}$ are the components of $\mathbf{a}$ itself.

6.4. What kind of object is a partial derivative?

This is worth stating carefully, because it is what the next section is built on.

At each point $(x, y) \in \mathbb{R}^2$, the limit (4) produces a single number. So $\partial f / \partial x$ assigns one real number to each point of the plane — which is to say it is itself a function

$$\frac{\partial f}{\partial x} : \mathbb{R}^2 \longrightarrow \mathbb{R},$$

of exactly the same type as f . Differentiating with respect to one variable does not reduce the number of variables; the results above are visibly still functions of both x and y .

And there are two of them.

7. The gradient

7.1. Collecting the partials

So a single function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ does not have *a* derivative. It has two — one per coordinate direction — and each is a function $\mathbb{R}^2 \rightarrow \mathbb{R}$ in its own right. Naming and carrying them separately would obscure the fact that they belong together: they are two answers to the same question, asked in two directions.

From the vector unit. The vector unit's opening move was exactly this. Faced with several related numbers, do not track them separately — package them into a single object of $\mathbb{R}^2$ and work with that. Position, velocity and force were treated that way.

Apply that move here.

Definition 7.1 (Gradient). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ have both partial derivatives at $\mathbf{x} = (x, y)$. The **gradient** of f at $\mathbf{x}$ is the vector in $\mathbb{R}^2$ whose components are the two partial derivatives:

$$\nabla f(\mathbf{x}) = \left[\frac{\partial f}{\partial x}(x, y), \frac{\partial f}{\partial y}(x, y) \right].$$

The symbol ∇ is read “del” or “nabla”. Note the signature that results:

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad \text{but} \quad \nabla f : \mathbb{R}^2 \rightarrow \mathbb{R}^2.$$

The gradient takes a point and returns a *vector*. This is a genuine departure from the one-variable case, where f' had the same signature as f .

7.2. Examples

Example 7.1. For $f(x, y) = x^2y + 3y^2$, the partials were computed in §6.3, so

$$\nabla f(x, y) = [2xy, x^2 + 6y].$$

At the point $(1, 2)$ this evaluates to $\nabla f(1, 2) = [4, 13]$ — a vector in $\mathbb{R}^2$ attached to that particular point. At a different point it is a different vector: $\nabla f(0, 1) = [0, 6]$, and $\nabla f(-1, 0) = [0, 1]$.

Example 7.2. For $f(x, y) = x^2 + y^2$,

$$\nabla f(x, y) = [2x, 2y] = 2[x, y].$$

At every point the gradient is twice the position vector, so it points directly away from the origin — which is where f is smallest. That is the first hint of §8.

Example 7.3. For $f(x, y) = \sin(xy)$, the partials were $y \cos(xy)$ and $x \cos(xy)$, so

$$\nabla f(x, y) = \cos(xy) [y, x].$$

Here the gradient happens to factor: a scalar depending on the point, times a vector. At $(\pi, \frac{1}{2})$, $\cos(\pi/2) = 0$, so $\nabla f = [0, 0]$ — the gradient is the zero vector there, and f is momentarily flat in *both* coordinate directions.

Example 7.4. For $f(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle$, the last example of §6.3 gave $\partial f / \partial x = a_1$ and $\partial f / \partial y = a_2$. Assembling them,

$$\nabla f(\mathbf{x}) = [a_1, a_2] = \mathbf{a}.$$

The gradient is $\mathbf{a}$ itself, the same at every point — the two-variable counterpart of the fact that a line of slope m has derivative m everywhere.

Remark (Why this is the right package). Collecting the two partials into a vector could look like mere bookkeeping. It is more than that, for a reason visible in the last example: the gradient of $\langle \mathbf{a}, \mathbf{x} \rangle$ came out as $\mathbf{a}$, not as two unrelated numbers that happen to be stored together. The package has an identity of its own. And because $\nabla f(\mathbf{x})$ is a vector in $\mathbb{R}^2$, everything the vector unit built applies to it: it has a length, it has a direction, it can be dotted with other vectors, it can be normalised, and it can be tested for orthogonality. The next section spends every one of those.

8. The directional derivative, and steepest ascent

§5.3 left a question open: the rate of change of f at a point is not well-posed until a direction is named, and §6 named only two of them. This section names all of them at once, and then asks which one is best.

8.1. The idea

Stand at a point $\mathbf{x}$ of the plane and pick a direction to walk in. A direction is a vector $\mathbf{u}$, and the natural question is: *as I set off that way, how fast does f change?*

That number depends on two things, and on nothing else: the point you are standing at, and the direction you chose. Give it a name.

The **directional derivative** $D_{\mathbf{u}}f(\mathbf{x})$ is the rate of change of f at the point $\mathbf{x}$, in the direction of the vector $\mathbf{u}$.

Said as a limit, it is (1) once more, with the step $h\mathbf{u}$ — a step of size h taken *in the direction* $\mathbf{u}$:

$$D_{\mathbf{u}}f(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{u}) - f(\mathbf{x})}{h}. \quad (6)$$

Every definition so far is a special case. Taking $\mathbf{u} = [1, 0]$ makes $\mathbf{x} + h\mathbf{u} = (x + h, y)$, and (6) becomes (4). Taking $\mathbf{u} = [0, 1]$ gives (5). The two partial derivatives are the directional derivative in the two directions we happened to name first.

A common confusion (Take $\mathbf{u}$ to have length one). $D_{\mathbf{u}}f$ is meant to measure a rate *per unit of distance travelled*, so the direction vector must be a unit vector — otherwise doubling the length of $\mathbf{u}$ doubles the answer without any change of direction, and the number measures the arrow rather than the hillside. Whenever a direction is given as a vector that is not already of length one, normalise it first: $\mathbf{u} = \mathbf{v} / \|\mathbf{v}\|$, as in §12.1 of the vector unit. Both $[1, 0]$ and $[0, 1]$ above are unit vectors, as they had to be.

8.2. Computing it from the gradient

The limit (6) is no more pleasant to evaluate than any other, and it does not have to be. The gradient already holds the answer.

Walking in the direction $\mathbf{u} = [u_1, u_2]$ means changing x at rate u_1 and changing y at rate u_2 , at the same time. Each partial derivative is the exchange rate for its own coordinate — $\partial f/\partial x$ says how much f moves per unit of x — so the contribution from the x -motion is $\frac{\partial f}{\partial x}u_1$, the contribution from the y -motion is $\frac{\partial f}{\partial y}u_2$, and the total rate is their sum:

$$D_{\mathbf{u}}f(\mathbf{x}) = \frac{\partial f}{\partial x}(\mathbf{x})u_1 + \frac{\partial f}{\partial y}(\mathbf{x})u_2.$$

That sum of products of matching components is a dot product, and the vector it is taken against is the gradient.

For a unit vector $\mathbf{u}$,

$$D_{\mathbf{u}}f(\mathbf{x}) = \langle \nabla f(\mathbf{x}), \mathbf{u} \rangle.$$

This is the first place the gradient pays for itself. The two partials answered two questions; packaged as a vector and dotted against $\mathbf{u}$, they answer all of them.

Remark. The argument just given is a motivation, not a proof — it assumes that the separate coordinate effects simply add, which is the content of the linear approximation of §9.2. Taking that formula as the definition of how f behaves near $\mathbf{x}$, the identity above is immediate; §9.2 closes the loop.

Example 8.1. Let $f(x, y) = x^2y + 3y^2$ and take the point $(1, 2)$, where §7.2 gave $\nabla f(1, 2) = [4, 13]$. Find the rate of change of f there in the direction of $\mathbf{v} = [3, 4]$.

First normalise: $\|\mathbf{v}\| = \sqrt{9 + 16} = 5$, so $\mathbf{u} = \frac{1}{5}[3, 4] = [\frac{3}{5}, \frac{4}{5}]$. Then

$$D_{\mathbf{u}}f(1, 2) = \langle [4, 13], [\frac{3}{5}, \frac{4}{5}] \rangle = \frac{12}{5} + \frac{52}{5} = \frac{64}{5} = 12.8.$$

Positive, so f increases as you set off that way.

Example 8.2. Same f and same point, but in the direction $\mathbf{v} = [-13, 4]$. Here $\|\mathbf{v}\| = \sqrt{169 + 16} = \sqrt{185}$, and

$$D_{\mathbf{u}}f(1, 2) = \frac{1}{\sqrt{185}} \langle [4, 13], [-13, 4] \rangle = \frac{-52 + 52}{\sqrt{185}} = 0.$$

The rate is zero: walking that way, f does not change at all to first order. By the orthogonality test of the vector unit's §11.7, that happened precisely because $[-13, 4]$ is orthogonal to the gradient — and the directions orthogonal to ∇f are the ones along which f holds still. Those are the level curves.

8.3. In which direction is the rate of change greatest?

Now the question worth asking. Standing at $\mathbf{x}$, every unit vector $\mathbf{u}$ gives some rate $D_{\mathbf{u}}f(\mathbf{x})$. Which $\mathbf{u}$ makes it as large as possible?

Write the formula of §8.2 and then rewrite the dot product the way the vector unit's §11.3 does, in terms of the angle θ between $\nabla f(\mathbf{x})$ and $\mathbf{u}$:

$$D_{\mathbf{u}}f(\mathbf{x}) = \langle \nabla f(\mathbf{x}), \mathbf{u} \rangle = \|\nabla f(\mathbf{x})\| \|\mathbf{u}\| \cos \theta = \|\nabla f(\mathbf{x})\| \cos \theta, \quad (7)$$

the last step because $\|\mathbf{u}\| = 1$.

Read what (7) says. As $\mathbf{u}$ swings around the point, the factor $\|\nabla f(\mathbf{x})\|$ does not move — it is fixed by the point, not by the direction. The *only* thing that varies is $\cos \theta$, and $\cos \theta$ is largest, equal to 1, exactly when $\theta = 0$. So the rate is greatest when the angle between $\mathbf{u}$ and the gradient is zero: when $\mathbf{u}$ points the same way as $\nabla f(\mathbf{x})$.

There is exactly one unit vector that does that. Assuming $\nabla f(\mathbf{x}) \neq \mathbf{0}$, take

$$\mathbf{u} = \frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|},$$

the normalisation of the gradient (vector unit §12.1). Substituting it into the formula of §8.2 confirms the value:

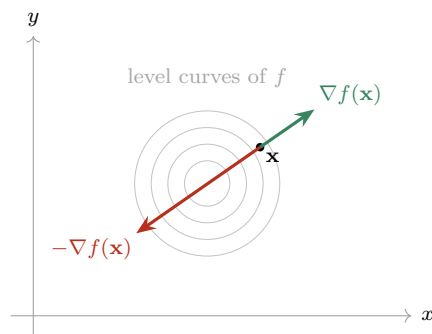
$$D_{\mathbf{u}}f(\mathbf{x}) = \left\langle \nabla f(\mathbf{x}), \frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|} \right\rangle = \frac{\langle \nabla f(\mathbf{x}), \nabla f(\mathbf{x}) \rangle}{\|\nabla f(\mathbf{x})\|} = \frac{\|\nabla f(\mathbf{x})\|^2}{\|\nabla f(\mathbf{x})\|} = \|\nabla f(\mathbf{x})\|,$$

using $\langle \mathbf{w}, \mathbf{w} \rangle = \|\mathbf{w}\|^2$ from the vector unit's §11.6.

So the rate of change at $\mathbf{x}$ is maximal along the direction of $\nabla f(\mathbf{x})$, and the maximal rate is $\|\nabla f(\mathbf{x})\|$. Since that value is positive, f genuinely increases in that direction: the gradient is an *ascent* direction, and no direction ascends faster.

At a point $\mathbf{x}$ where both partial derivatives exist and $\nabla f(\mathbf{x}) \neq \mathbf{0}$, the gradient $\nabla f(\mathbf{x})$ points in the direction of **steepest ascent** of f , and its length $\|\nabla f(\mathbf{x})\|$ is the rate of climb in that direction.

Remark (The two extremes, and the flat directions). The same reading of (7) settles the other cases at no extra cost. $\cos \theta$ is smallest, equal to -1 , when $\theta = \pi$ — so the rate is *most negative* when $\mathbf{u}$ points opposite to the gradient, and $-\nabla f(\mathbf{x})$ is the direction of **steepest descent**. And $\cos \theta = 0$ when $\theta = \pi/2$, so the rate is zero exactly along the directions orthogonal to the gradient, which is what the second example of §8.2 found by arithmetic. Those are the directions in which f holds still: the gradient crosses the level curves of f at right angles.



The two arrows show *directions*; their drawn lengths are not to scale. The gradient heads toward the level curves where f is larger. Reversing it gives the direction of steepest descent — here, straight toward the low point at the centre.

Remark (The same thing happens in one variable). With one variable there are only two directions to move, and the gradient is the single number $f'(x)$. If $f'(x) > 0$ then f increases as x increases, so “uphill” is the positive direction; if $f'(x) < 0$, uphill is the negative direction. In both cases uphill is the direction the derivative points, and $|f'(x)|$ says how steep the slope is. The gradient is that statement with more than two directions available — which is why it must be a vector rather than a sign.

8.4. Gradient descent

The descent half of the last remark is the more useful one in practice, because most problems worth solving are phrased as *make this quantity small*.

Suppose $f(x, y)$ measures how badly something is performing — the error of a prediction, the cost of a design — and you want the (x, y) that makes it as small as possible. You cannot see the whole surface. But at whatever point you are standing, you can compute ∇f , and $-\nabla f$ tells you which way is downhill, faster than any other way. So: step that way, and repeat.

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k), \quad \alpha > 0.$$

The number α is the **step size**, controlling how far to move before recomputing. This is **gradient descent**, and it is the workhorse of modern optimisation and machine learning — the method by which very nearly every neural network in existence is trained. It exists because of the boxed statement in §8.3 and nothing more.

Looking ahead. Two questions are already visible. The iteration stops making progress where $\nabla f = \mathbf{0}$, so those points deserve a name and a study — they are the candidates for maxima and minima. And knowing only the slope, gradient descent has no idea whether the ground ahead curves up or down, so it cannot tell how big a step is safe; answering that needs a *second* derivative, which in more than one variable is neither a number nor a vector. Part 2 of this unit takes both up.

9. Linear approximation

One of the oldest uses of a derivative is to replace a complicated function, near a single point, by a simple one. The gradient does this in the plane exactly as f' does on the line — and comparing the two formulas shows what kind of object the gradient really is.

9.1. One variable

Return to the picture in §1.1. The tangent line at $x = a$ touches the graph at that point and has slope $f'(a)$, so its equation is

$$L(x) = f(a) + f'(a)(x - a). \quad (8)$$

Near a the curve and its tangent line are hard to tell apart, so L can stand in for f :

$$f(x) \approx f(a) + f'(a)(x - a) \quad \text{for } x \text{ near } a.$$

This is the **linear approximation** (or *linearisation*) of f at a . Read (8) in three pieces: start from the known value $f(a)$, measure how far you have moved with $x - a$, and convert that displacement into a change in f by multiplying by the rate $f'(a)$.

Example 9.1. Estimate $\sqrt{4.1}$ without a calculator. Take $f(x) = \sqrt{x}$ and $a = 4$, where the value is known exactly. Then $f(4) = 2$ and $f'(x) = \frac{1}{2\sqrt{x}}$, so $f'(4) = \frac{1}{4}$. Hence

$$\sqrt{4.1} \approx 2 + \frac{1}{4}(4.1 - 4) = 2 + \frac{0.1}{4} = 2.025.$$

The true value is $2.02484\dots$, so the estimate is off by about 1.5×10^{-4} — from arithmetic done in one line.

9.2. Two variables

Now let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and let $\mathbf{a} = (a_1, a_2)$ be the point we are near. Every piece of (8) survives, but two of them change type.

The starting value $f(\mathbf{a})$ is still a scalar. The displacement is now $\mathbf{x} - \mathbf{a}$, a *vector* — it must record how far we moved in both coordinates, not just one. And the rate is now $\nabla f(\mathbf{a})$, also a vector. To turn a vector of rates and a vector of displacements into a single number, there is exactly one natural operation available, and the vector unit built it: the dot product.

Definition 9.1 (Linear approximation in two variables). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ have a gradient at $\mathbf{a}$. The **linear approximation** of f at $\mathbf{a}$ is

$$L(\mathbf{x}) = f(\mathbf{a}) + \langle \nabla f(\mathbf{a}), \mathbf{x} - \mathbf{a} \rangle, \quad (9)$$

and $f(\mathbf{x}) \approx L(\mathbf{x})$ for $\mathbf{x}$ near $\mathbf{a}$.

Set the two formulas side by side:

$$\underbrace{f(a) + f'(a)(x - a)}_{\text{one variable}} \quad \longleftrightarrow \quad \underbrace{f(\mathbf{a}) + \langle \nabla f(\mathbf{a}), \mathbf{x} - \mathbf{a} \rangle}_{\text{two variables}}.$$

They are the same formula. The only change is that a product of two numbers has become a dot product of two vectors.

Written out in coordinates, (9) reads

$$L(x, y) = f(a_1, a_2) + \frac{\partial f}{\partial x}(\mathbf{a})(x - a_1) + \frac{\partial f}{\partial y}(\mathbf{a})(y - a_2),$$

one term per direction: each partial derivative converts the displacement in its own coordinate into a contribution to the change in f , and the two contributions are added.

Example 9.2. Let $f(x, y) = x^2 + y^2$ and approximate near $\mathbf{a} = (1, 2)$. There $f(\mathbf{a}) = 1 + 4 = 5$, and from §7.2, $\nabla f(x, y) = [2x, 2y]$, so $\nabla f(\mathbf{a}) = [2, 4]$. Then

$$L(x, y) = 5 + \langle [2, 4], [x - 1, y - 2] \rangle = 5 + 2(x - 1) + 4(y - 2).$$

At $(1.1, 1.9)$ this predicts

$$L(1.1, 1.9) = 5 + 2(0.1) + 4(-0.1) = 5 + 0.2 - 0.4 = 4.8,$$

against the true value $1.21 + 3.61 = 4.82$. Note the two terms pulling opposite ways: moving right raises f , moving down lowers it, and the approximation adds both effects.

Remark (What the graph of L is). For one variable the linear approximation is a tangent *line*. For two it is a tangent *plane*, resting on the surface at a single point. In both cases it is the flat object that best matches f at $\mathbf{a}$ — and it is the same plane whose slices, in the two coordinate directions, are the tangent lines drawn in §6.2.

Remark (Why the gradient deserves to be called the derivative). (9) is the strongest answer to the question of what replaces f' in two variables. In one variable, $f'(a)$ is the number that makes the best linear approximation work. In two, $\nabla f(\mathbf{a})$ is the vector that plays that role — the same job, the same position in the formula, a different type of object. Collecting the partials into a vector was not bookkeeping after all: it produced the thing that behaves like a derivative.

Remark (And it closes the loop of §8.2). Put $\mathbf{x} = \mathbf{a} + h\mathbf{u}$ into (9), so that the displacement is $h\mathbf{u}$. The dot product is linear in that slot, so

$$L(\mathbf{a} + h\mathbf{u}) = f(\mathbf{a}) + h \langle \nabla f(\mathbf{a}), \mathbf{u} \rangle,$$

and the change in the approximation, divided by the distance h travelled, is $\langle \nabla f(\mathbf{a}), \mathbf{u} \rangle$ — exactly the directional derivative claimed in §8.2. The formula asserted there is the linear approximation read along a single direction.

Looking ahead. Two roads lead out of here. Asking where L is flat — where $\nabla f(\mathbf{a}) = \mathbf{0}$ — locates the candidates for maxima and minima, the points at which gradient descent stalls. And approximating with something better than a flat object requires a *second* derivative, which for a function of two variables turns out to be neither a number nor a vector but a *matrix*. Both belong to Part 2, which also lifts everything in §5–§9 from the plane to $\mathbb{R}^n$.